

Qualification Pack



Automotive IIOT Application Specialist

QP Code: ASC/Q6415

Version: 1.0

NSQF Level: 6

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ASC/Q6415: Automotive IIOT Application Specialist

Brief Job Description

The individual at this job is responsible for managing and monitoring activities of integration of machines, robots, and Automation Systems in Healthy Communication, establish healthy communication using network protocols, remote monitoring and fetch vital machine data using IIOT edge devices within an organization for all its processes, the new development, production and application phases.

Personal Attributes

This job requires the individual to work at a desk-based job for long hours. The individual should be result oriented and should also be able to demonstrate skills for information ordering, analytical reasoning, problem solving, time management, oral expression and comprehension.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N9810: Manage work and resources \(Manufacturing\)](#)
2. [ASC/N6438: Plan and design a network architecture using IIOT Solutions](#)
3. [ASC/N6447: Manage integration of machineries / Automation Systems in Healthy Communication via IIOT Sensor](#)
4. [ASC/N6448: Use Analytics, Edge & Cloud Computing Technologies to make Predictions on Production & Machineries](#)
5. [DGT/VSQ/N0104: Employability Skills \(120 Hours\)](#)
6. [ASC/N8312: Liaison with vendors and other departments](#)

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
Country	India
NSQF Level	6

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Credits	21
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2151.0602
Minimum Educational Qualification & Experience	B.E./B.Tech (In relevant field) with 1 Year of experience relevant OR M.Tech (Persuing 2nd Year) OR Certificate-NSQF (Automotive IIOT Application Engineer Level 5) with 2 Years of experience relevant
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	23/06/2026
NSQC Approval Date	23/06/2023
Version	1.0
Reference code on NQR	QG-06-AU-00572-2023-V1-ASDC
NQR Version	1.1

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ASC/N9810: Manage work and resources (Manufacturing)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Maintain Health and Hygiene
- Effective waste management practices
- Material/energy conservation practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** implement safe working practices for dealing with hazards to ensure safety of self and others
- PC3.** conduct regular checks of the machines with support of the maintenance team to identify potential hazards
- PC4.** ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions
- PC5.** organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices
- PC6.** fill daily check sheet to report improvements done and risks identified
- PC7.** ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others
- PC8.** report any identified breaches in health, safety and security policies and procedures to the designated person

Maintain Health and Hygiene

To be competent, the user/individual on the job must be able to:

- PC9.** ensure workplace, equipment, restrooms etc. are sanitized regularly
- PC10.** ensure team is aware about hygiene and sanitation regulations and following them on the shop floor
- PC11.** ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace
- PC12.** report advanced hygiene and sanitation issues to appropriate authority
- PC13.** follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc
- PC14.** wear and dispose PPEs regularly and appropriately

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Effective waste management practices

To be competent, the user/individual on the job must be able to:

PC15. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP

PC16. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste

Material/energy conservation practices

To be competent, the user/individual on the job must be able to:

PC17. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively

PC18. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water

PC19. identify possibilities of using renewable energy and environment friendly fuels

PC20. identify processes where material and energy/electricity utilization can be optimized

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organisation procedures for health, safety and security, individual role and responsibilities in this context

KU2. the organisation's emergency procedures for different emergency situations and the importance of following the same

KU3. evacuation procedures for workers and visitors

KU4. how and when to report hazards as well as the limits of responsibility for dealing with hazards

KU5. potential hazards, risks and threats based on the nature of work

KU6. various types of fire extinguisher

KU7. various types of safety signs and their meaning

KU8. appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

KU9. relevant standards, procedures and policies related to 5S followed in the company

KU10. the various materials used and their storage norms

KU11. importance of efficient utilisation of material and water

KU12. basics of electricity and prevalent energy efficient devices

KU13. common practices of conserving electricity

KU14. common sources and ways to minimize pollution

KU15. categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

KU16. waste management techniques

KU17. significance of greening

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	20	13	-	8
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	4	2	-	2
PC2. implement safe working practices for dealing with hazards to ensure safety of self and others	3	1	-	2
PC3. conduct regular checks of the machines with support of the maintenance team to identify potential hazards	2	2	-	1
PC4. ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions	3	2	-	1
PC5. organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices	2	-	-	-
PC6. fill daily check sheet to report improvements done and risks identified	2	2	-	-
PC7. ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others	2	2	-	1
PC8. report any identified breaches in health, safety and security policies and procedures to the designated person	2	2	-	1
<i>Maintain Health and Hygiene</i>	13	7	-	5
PC9. ensure workplace, equipment, restrooms etc. are sanitized regularly	3	2	-	1
PC10. ensure team is aware about hygiene and sanitation regulations and following them on the shop floor	2	1	-	-
PC11. ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. report advanced hygiene and sanitation issues to appropriate authority	1	1	-	1
PC13. follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc	2	1	-	1
PC14. wear and dispose PPEs regularly and appropriately	3	-	-	1
<i>Effective waste management practices</i>	6	4	-	1
PC15. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP	3	2	-	-
PC16. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste	3	2	-	1
<i>Material/energy conservation practices</i>	11	6	-	6
PC17. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively	2	2	-	1
PC18. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water	3	2	-	1
PC19. identify possibilities of using renewable energy and environment friendly fuels	3	1	-	2
PC20. identify processes where material and energy/electricity utilization can be optimized	3	1	-	2
NOS Total	50	30	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9810
NOS Name	Manage work and resources (Manufacturing)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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ASC/N6438: Plan and design a network architecture using IIOT Solutions

Description

This unit is about designing a network having machines, robot, automation systems, and network dashboards while taking various considerations, regulations, and Pertinent requirements into account

Scope

The scope covers the following :

- Prepare for building architecture
- Interpret the various network parameters required as per design
- Develop the network as per design requirements
- Monitor the systems and devices mapping in network

Elements and Performance Criteria

Prepare for building architecture

To be competent, the user/individual on the job must be able to:

- PC1.** Obtain the design document and project develop requirements from the organization/OEM
- PC2.** Identify and select appropriate development tools, coding language, development platform, OS etc. as per the project requirements
- PC3.** Evaluate the IIOT network requirement like node allotment, station setting
- PC4.** Identify and select the machines, automation system to be connected in the IIOT network
- PC5.** Identify the suitable communication model, topology to connect the machines & automation system

Interpret the various network parameters required as per design

To be competent, the user/individual on the job must be able to:

- PC6.** Support in designing of wireless/wired network nodes while taking into consideration the varieties of IIOT sensors, I/O link, edge devices, cloud service, and other networking devices
- PC7.** Ensure appropriate wired/wireless connectivity protocols are applied for Machine-to-Machine communications
- PC8.** Ensure appropriate wired/wireless connectivity protocols are applied for edge device-cloud communications
- PC9.** Verify the network parameters like communication speed, network topology, node address of different devices in the IIOT network

Develop the network as per design requirements

To be competent, the user/individual on the job must be able to:

- PC10.** Support team in building code and configure software, its components and their oriented design
- PC11.** Support team in configuring ports and interfaces, Sender-Receiver communication and Client-Server communication as per the requirement
- PC12.** Guide the team during assigning nodes to every device in the system
- PC13.** Monitor that the nodes are optimized appropriately as per design document

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- PC14.** Validate codes of all the components of architecture to ensure required output
 - PC15.** Approve network design based on industrial requirement
 - PC16.** Evaluate regulatory aspects of shopfloor network such as permitted frequency bands, possible interference due to high voltage line
 - PC17.** Design and develop networking dashboards used for network monitoring with help from edge computing devices
 - PC18.** Monitor timelines/deadlines and host review meetings
 - PC19.** Ensure that the team is documenting the final network architecture plan appropriately
- Monitor the systems and devices mapping in network*
- To be competent, the user/individual on the job must be able to:
- PC20.** Support the team during selection of the devices & checking their compatibility with the network and system
 - PC21.** Monitor the IIOT network parameters like node address, communication speed and their effect on the network
 - PC22.** Monitor that the team is mapping the node address of devices connected in the IIOT network appropriately and as per the design document and organizational standards
 - PC23.** Verify that defined security standards and encryptions are followed in the EDGE boards and DATA links and as per the design document
 - PC24.** Support the team in conducting communication test across all devices in the IIOT Network
 - PC25.** Record the healthy status of all devices & report to organizational project review committee

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** Use of robots in different manufacturing and assembly operations
- KU3.** Safe operation of electronic equipment like computers, robotic systems etc
- KU4.** Relevant standards, procedures and policies related to robotic operations followed in the company
- KU5.** Data safety and Non-Discloser's Norms
- KU6.** Designing and developing networks for the solution
- KU7.** Cyber safety and Work Confidentiality good practices
- KU8.** Data and folder standards
- KU9.** Different check sheets and technical documents
- KU10.** IIOT sensors, I/O link, edge devices, cloud service, and other networking devices required
- KU11.** Automation Systems Like PLC, Pneumatics, Hydraulics, HMI, Industrial Sensors
- KU12.** Wired/wireless connectivity protocols for Machine-to-Machine communications
- KU13.** Wired/wireless connectivity protocols for edge device-cloud communications
- KU14.** Network parameters like communication speed, network topology, node address of different devices
- KU15.** Impacts of network on the environment and human health

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write basic level notes and observations
- GS2.** Modify work practices to improve them
- GS3.** Note down observations related to the process
- GS4.** Write information documents to internal departments and internal teams
- GS5.** Monitoring of tasks completion efficiently and accurately within stipulated time
- GS6.** Persuade team members to follow all guidelines and improvement procedures
- GS7.** Make timely decisions for efficient utilization of resources
- GS8.** Be punctual and utilize time efficiently
- GS9.** Follow shop floor rules and avoid deviations
- GS10.** Encourage self and other to take greater responsibilities in the task
- GS11.** Question the process head in order to understand the activity
- GS12.** Do what is right, not what is a popular practice
- GS13.** Lead by example in plant premises and work area
- GS14.** Ensure self-discipline and cleanliness on daily basis
- GS15.** Use common sense and knowledge in day-to-day decision making
- GS16.** Accept additional responsibilities to provide more flexibility to the team
- GS17.** Use reasoning skills in identifying and resolving problems in industrial automation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for building architecture</i>	9	9	-	8
PC1. Obtain the design document and project develop requirements from the organization/OEM	1	1	-	1
PC2. Identify and select appropriate development tools, coding language, development platform, OS etc. as per the project requirements	3	2	-	2
PC3. Evaluate the IIOT network requirement like node allotment, station setting	2	2	-	2
PC4. Identify and select the machines, automation system to be connected in the IIOT network	2	2	-	2
PC5. Identify the suitable communication model, topology to connect the machines & automation system	1	2	-	1
<i>Interpret the various network parameters required as per design</i>	6	6	-	4
PC6. Support in designing of wireless/wired network nodes while taking into consideration the varieties of IIOT sensors, I/O link, edge devices, cloud service, and other networking devices	2	2	-	1
PC7. Ensure appropriate wired/wireless connectivity protocols are applied for Machine-to-Machine communications	1	1	-	1
PC8. Ensure appropriate wired/wireless connectivity protocols are applied for edge device-cloud communications	1	1	-	1
PC9. Verify the network parameters like communication speed, network topology, node address of different devices in the IIOT network	2	2	-	1
<i>Develop the network as per design requirements</i>	17	19	-	6
PC10. Support team in building code and configure software, its components and their oriented design	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Support team in configuring ports and interfaces, Sender-Receiver communication and Client-Server communication as per the requirement	2	2	-	1
PC12. Guide the team during assigning nodes to every device in the system	1	2	-	1
PC13. Monitor that the nodes are optimized appropriately as per design document	2	2	-	-
PC14. Validate codes of all the components of architecture to ensure required output	2	2	-	1
PC15. Approve network design based on industrial requirement	1	1	-	-
PC16. Evaluate regulatory aspects of shopfloor network such as permitted frequency bands, possible interference due to high voltage line	2	2	-	1
PC17. Design and develop networking dashboards used for network monitoring with help from edge computing devices	3	3	-	1
PC18. Monitor timelines/deadlines and host review meetings	1	2	-	-
PC19. Ensure that the team is documenting the final network architecture plan appropriately	1	1	-	-
<i>Monitor the systems and devices mapping in network</i>	8	6	-	2
PC20. Support the team during selection of the devices & checking their compatibility with the network and system	1	1	-	1
PC21. Monitor the IIOT network parameters like node address, communication speed and their effect on the network	2	1	-	-
PC22. Monitor that the team is mapping the node address of devices connected in the IIOT network appropriately and as per the design document and organizational standards	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. Verify that defined security standards and encryptions are followed in the EDGE boards and DATA links and as per the design document	2	1	-	1
PC24. Support the team in conducting communication test across all devices in the IIOT Network	1	1	-	-
PC25. Record the healthy status of all devices & report to organizational project review committee	1	1	-	-
NOS Total	40	40	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6438
NOS Name	Plan and design a network architecture using IIOT Solutions
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

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ASC/N6447: Manage integration of machineries / Automation Systems in Healthy Communication via IIOT Sensor

Description

This NOS unit is about performing task related to integration of machines and automation systems using industrial networking protocols, IIOT devices used in manufacturing processes to meet the specification set by the organization.

Scope

The scope covers the following :

- Support in installation of the elements in different layers of industrial network architecture & protocols
- Establish communication between systems using IIOT Sensors, I/O link master and IIOT EDGE Computing Devices
- Ensure IIOT Network security among machineries & automation systems

Elements and Performance Criteria

Support in installation of the elements in different layers of industrial network architecture & protocols

To be competent, the user/individual on the job must be able to:

- PC1.** Prepare an integration plan to map devices and systems as per network design document
- PC2.** Interpret the installed machines, automation elements, system, and robots into different layers of network architecture like field devices, control devices as per network design
- PC3.** Verify the network consists of devices, automation system and robots
- PC4.** Guide the team to select suitable network protocols like MODBUS, CC-LINK, Profinet, Profibus, OPC UA, MQTT etc. based on control system requirements
- PC5.** Ensure that team has connected the intelligent devices and system using suitable network topology like STAR, LINE, RING as per network design document
- PC6.** Monitor the activities related to installation of the systems on the pre-planned locations to distribute the network for max connectivity and reach
- PC7.** Verify the systems & devices integrated as per design documents
- PC8.** Guide the team during verification of the connections of IIoT Sensors, I/O-Links to the machines, robots Automation systems with the appropriate Network Protocol Like OPC UA, Mod Bus

Establish communication between systems using IIOT Sensors, I/O Link Master & IIOT EDGE Computing Devices

To be competent, the user/individual on the job must be able to:

- PC9.** Ensure that the automation elements like sensors, control devices to I/O link master are connected as per SOP
- PC10.** Establish the communication between automation system, machines, and robots by doing parameter setting like baud rate, distance, node ID and node type

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- PC11.** Support the team in setting of the network parameters of automation system on the device manufacturers software as per SOP and organizational guidelines
- PC12.** Verify the Node-wise Communication having IIOT sensors, edge computing devices
- PC13.** Initiate the communication between Nodes in the IIOT Network like Machine-to-Machine Communication and ensure healthy communication between them
- PC14.** Configure the EDGE devices as per the IIOT network configuration
- PC15.** Guide the team to establish healthy communication between machines/devices and check for device duplicate
- PC16.** Support the team during testing of the functioning of backup systems, power connectivity and security access to the EDGE devices from both online and offline mode
- PC17.** Manage the activities related to server allotment and database for cloud computing

Ensure IIOT Network security among machineries & automation systems

To be competent, the user/individual on the job must be able to:

- PC18.** Ensure physical security of the network contains IIOT Edge Devices, IIOT Sensors, Machines, Robots and Automation System
- PC19.** Guide the team to verify the physical security and environment for the hardware for hasslefree operations
- PC20.** Monitor that the network is protected from unauthorized access or malicious internet
- PC21.** Ensure only authorized devices should be able to connect to the network

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** Software and 3D tools used in organisation.
- KU3.** Network Security Protocols
- KU4.** Safe operation of electronic equipment like computers, robotic systems etc
- KU5.** Relevant standards, procedures and policies related to robotic operations followed in the company
- KU6.** Data safety and Non Disclosure Norms
- KU7.** Cyber safety and Work Confidentiality good practices
- KU8.** E-Plan as per Customer Requirement
- KU9.** Team cohesion and collaborative working
- KU10.** IIOT Sensors Connection Details

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write basic level notes and observations
- GS2.** Modify work practices to improve them

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- GS3.** Note down observations related to the process
- GS4.** Write information documents to internal departments and internal teams
- GS5.** Monitoring of tasks completion efficiently and accurately within stipulated time
- GS6.** Persuade team members to follow all guidelines and improvement procedures
- GS7.** Make timely decisions for efficient utilization of resources
- GS8.** Be punctual and utilize time efficiently
- GS9.** Follow shop floor rules and avoid deviations
- GS10.** Encourage self and other to take greater responsibilities in the task
- GS11.** Question the process head in order to understand the activity
- GS12.** Do what is right, not what is a popular practice
- GS13.** Lead by example in plant premises and work area
- GS14.** Ensure self-discipline and cleanliness on daily basis.
- GS15.** Use common sense and knowledge in day-to-day decision making
- GS16.** Accept additional responsibilities to provide more flexibility to the team
- GS17.** Use reasoning skills in identifying and resolving problems in industrial automation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Support in installation of the elements in different layers of industrial network architecture & protocols</i>	20	16	-	8
PC1. Prepare an integration plan to map devices and systems as per network design document	2	1	-	1
PC2. Interpret the installed machines, automation elements, system, and robots into different layers of network architecture like field devices, control devices as per network design	3	1	-	1
PC3. Verify the network consists of devices, automation system and robots	3	2	-	1
PC4. Guide the team to select suitable network protocols like MODBUS, CC-LINK, Profinet, Profibus, OPC UA, MQTT etc. based on control system requirements	3	1	-	1
PC5. Ensure that team has connected the intelligent devices and system using suitable network topology like STAR, LINE, RING as per network design document	2	3	-	1
PC6. Monitor the activities related to installation of the systems on the pre-planned locations to distribute the network for max connectivity and reach	3	3	-	1
PC7. Verify the systems & devices integrated as per design documents	2	3	-	1
PC8. Guide the team during verification of the connections of IIoT Sensors, I/O-Links to the machines, robots Automation systems with the appropriate Network Protocol Like OPC UA, Mod Bus	2	2	-	1
<i>Establish communication between systems using IIOT Sensors, I/O Link Master & IIOT EDGE Computing Devices</i>	15	17	-	8
PC9. Ensure that the automation elements like sensors, control devices to I/O link master are connected as per SOP	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Establish the communication between automation system, machines, and robots by doing parameter setting like baud rate, distance, node ID and node type	2	2	-	1
PC11. Support the team in setting of the network parameters of automation system on the device manufacturers software as per SOP and organizational guidelines	2	2	-	1
PC12. Verify the Node-wise Communication having IIOT sensors, edge computing devices	2	2	-	1
PC13. Initiate the communication between Nodes in the IIOT Network like Machine-to-Machine Communication and ensure healthy communication between them	3	2	-	1
PC14. Configure the EDGE devices as per the IIOT network configuration	1	1	-	1
PC15. Guide the team to establish healthy communication between machines/devices and check for device duplicate	1	2	-	-
PC16. Support the team during testing of the functioning of backup systems, power connectivity and security access to the EDGE devices from both online and offline mode	1	2	-	1
PC17. Manage the activities related to server allotment and database for cloud computing	1	2	-	1
<i>Ensure IIOT Network security among machineries & automation systems</i>	5	7	-	4
PC18. Ensure physical security of the network contains IIOT Edge Devices, IIOT Sensors, Machines, Robots and Automation System	1	2	-	1
PC19. Guide the team to verify the physical security and environment for the hardware for hasslefree operations	2	2	-	1
PC20. Monitor that the network is protected from unauthorized access or malicious internet	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Ensure only authorized devices should be able to connect to the network	1	1	-	1
NOS Total	40	40	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6447
NOS Name	Manage integration of machineries / Automation Systems in Healthy Communication via IIOT Sensor
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

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ASC/N6448: Use Analytics, Edge & Cloud Computing Technologies to make Predictions on Production & Machineries

Description

This NOS unit is about to perform prediction of Production Process, Machineries Using Analytics, Edge & Cloud Computing Technologies

Scope

The scope covers the following :

- Fetch the process and machine data using edge computing devices
- Use of statistical tools to organize and analyze the data
- Apply algorithm to make predictions using cloud computing technologies

Elements and Performance Criteria

Fetch the process and machine data using edge computing devices

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the relevant source of data from machines and process
- PC2.** Select the machine communication protocol as per design document
- PC3.** Check for duplicate data & validate it
- PC4.** Monitor the process using edge computing devices like production or machine status
- PC5.** Customize the edge computing device dashboard as per customer requirement

Use of statistical tools to organize and analyse the data

To be competent, the user/individual on the job must be able to:

- PC6.** Select appropriate statistical tools for data analysis
- PC7.** Apply the statistical tools like regression, standard deviation on the data to analyse the trend and patterns
- PC8.** Perform inferential or descriptive analysis on the data to extrapolate
- PC9.** Find correlation amongst the selected attributes of the data and statistical proposition

Apply algorithm to make predictions using cloud computing technologies

To be competent, the user/individual on the job must be able to:

- PC10.** Analyse the data and trends, then create a model on that using algorithm
- PC11.** Fetch the model into the cloud servers
- PC12.** Enable the client services to access data from cloud domain
- PC13.** Make predictions on production process, machine status by access the model from cloud domain

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Safe operation of electronic equipment like computers, robotic systems etc.
- KU2.** Relevant standards, procedures and policies related to robotic operations followed in the company
- KU3.** Data safety and Non-Disclosure Norms
- KU4.** Cyber safety and Work Confidentiality good practices
- KU5.** Data and folder standards
- KU6.** Statistical Tools to organize and analyze the data
- KU7.** Algorithm to make predictions using cloud computing technologies
- KU8.** Team cohesion and collaborative working
- KU9.** Types of Data Algorithm, Statistical Tools
- KU10.** Cloud Services-Servers, Clients, Network Protocols, Remote Servers

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Modify work practices to improve them
- GS2.** Note down observations related to the process
- GS3.** Write information documents to internal departments and internal teams
- GS4.** Monitoring of tasks completion efficiently and accurately within stipulated time
- GS5.** Persuade team members to follow all guidelines and improvement procedures
- GS6.** Make timely decisions for efficient utilization of resources
- GS7.** Be punctual and utilize time efficiently
- GS8.** Follow shop floor rules and avoid deviations
- GS9.** Encourage self and other to take greater responsibilities in the task
- GS10.** Question the process head in order to understand the activity
- GS11.** Do what is right, not what is a popular practice
- GS12.** Lead by example in plant premises and work area
- GS13.** Ensure self-discipline and cleanliness on daily basis
- GS14.** Use common sense and knowledge in day-to-day decision making
- GS15.** Accept additional responsibilities in order to provide more flexibility to the team
- GS16.** Use reasoning skills in identifying and resolving problems in industrial automation

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fetch the process and machine data using edge computing devices</i>	13	13	-	7
PC1. Identify the relevant source of data from machines and process	2	2	-	2
PC2. Select the machine communication protocol as per design document	3	3	-	2
PC3. Check for duplicate data & validate it	3	3	-	1
PC4. Monitor the process using edge computing devices like production or machine status	2	2	-	1
PC5. Customize the edge computing device dashboard as per customer requirement	3	3	-	1
<i>Use of statistical tools to organize and analyse the data</i>	14	14	-	7
PC6. Select appropriate statistical tools for data analysis	3	3	-	2
PC7. Apply the statistical tools like regression, standard deviation on the data to analyse the trend and patterns	4	4	-	2
PC8. Perform inferential or descriptive analysis on the data to extrapolate	4	4	-	2
PC9. Find correlation amongst the selected attributes of the data and statistical proposition	3	3	-	1
<i>Apply algorithm to make predictions using cloud computing technologies</i>	13	13	-	6
PC10. Analyse the data and trends, then create a model on that using algorithm	4	4	-	2
PC11. Fetch the model into the cloud servers	3	3	-	1
PC12. Enable the client services to access data from cloud domain	3	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Make predictions on production process, machine status by access the model from cloud domain	3	3	-	2
NOS Total	40	40	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6448
NOS Name	Use Analytics, Edge & Cloud Computing Technologies to make Predictions on Production & Machineries
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

DGT/VSQ/N0104: Employability Skills (120 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress
- PC5.** follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC6.** follow and promote environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC7.** recognize the significance of 21st Century Skills for employment

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- PC8.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC9.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC10.** use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts
- PC11.** speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front
- PC12.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC13.** write short messages, notes, letters, e-mails etc., using accurate English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC14.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC15.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC16.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC17.** use active listening techniques for effective communication
- PC18.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC19.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC20.** • ensure personal behaviour, conduct, and use appropriate communication by taking gender into consideration
- PC21.** empathize with a PwD and aid a PwD, if asked
- PC22.** escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC23.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC24.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC25.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC26.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

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To be competent, the user/individual on the job must be able to:

- PC27.** operate digital devices and use their features and applications securely and safely
- PC28.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC29.** display responsible online behaviour while using various social media platforms
- PC30.** create a personal email account, send and process received messages as per requirement
- PC31.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC32.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of Entrepreneurship and Enterprises
- PC34.** use research and networking skills to identify and assess opportunities for potential business
- PC35.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC36.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC37.** identify different types of customers
- PC38.** identify and respond to customer requests and needs in a professional manner
- PC39.** use appropriate tools to collect customer feedback
- PC40.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC41.** create a professional Curriculum vitae (Résumé)
- PC42.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC43.** apply to identified job openings using offline /online methods as per requirement
- PC44.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC45.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills
- KU2.** different learning and employability related portals
- KU3.** various constitutional and personal values
- KU4.** different environmentally sustainable practices and their importance
- KU5.** Twenty first (21st) century skills and their importance

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- KU6.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU7.** importance of career development and setting long- and short-term goals
- KU8.** Do's and don'ts of effective communication
- KU9.** POSH Act
- KU10.** inclusivity and its importance
- KU11.** different types of disabilities and appropriate verbal and non-verbal communication and behaviour towards PwD
- KU12.** different types of financial institutes, products, and services
- KU13.** components of salary and how to compute income and expenditure
- KU14.** importance of maintaining safety and security in offline and online financial transactions
- KU15.** different legal rights and laws
- KU16.** different types of digital devices and the procedure to operate them safely and securely
- KU17.** how to create and operate an e- mail account
- KU18.** use applications such as word processors, spreadsheets etc.
- KU19.** different types of Enterprises and ways to identify business opportunities
- KU20.** types and needs of customers
- KU21.** how to apply for a job and prepare for an interview
- KU22.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	2	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress	-	-	-	-
PC5. follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC6. follow and promote environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	3	-	-
PC7. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC8. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC9. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts	-	-	-	-
PC11. speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front	-	-	-	-
PC12. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC13. write short messages, notes, letters, e-mails etc., using accurate English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC14. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC15. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	3	-	-
PC16. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC17. use active listening techniques for effective communication	-	-	-	-
PC18. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC19. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC20. • ensure personal behaviour, conduct, and use appropriate communication by taking gender into • consideration	-	-	-	-
PC21. empathize with a PwD and aid a PwD, if asked	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC23. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC24. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-
PC25. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC26. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	2	3	-	-
PC27. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC28. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC29. display responsible online behaviour while using various social media platforms	-	-	-	-
PC30. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC31. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC32. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. identify different types of Entrepreneurship and Enterprises	-	-	-	-
PC34. use research and networking skills to identify and assess opportunities for potential business	-	-	-	-
PC35. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC36. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC37. identify different types of customers	-	-	-	-
PC38. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC39. use appropriate tools to collect customer feedback	-	-	-	-
PC40. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	4	-	-
PC41. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC42. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC43. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC44. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC45. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0104
NOS Name	Employability Skills (120 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

ASC/N8312: Liaison with vendors and other departments

Description

This NOS unit is about coordination with vendor organizations/departments and to run and improve the processes successfully.

Scope

The scope covers the following :

- Collaboration with system developers
- Pre and post support
- Process and technology study and scope of development

Elements and Performance Criteria

Collaboration with system developers

To be competent, the user/individual on the job must be able to:

- PC1.** interact with different vendors for developing the robotic automation system to meet all the information integration and information flow starting from conception of the idea till implementation of the robotic system
- PC2.** ensure that integrators/developers incorporate all the necessary requirement such what level of information access will be different department and different level of people as per requirement/hierarchy system
- PC3.** carry out the techno-commercial feasibility analysis with system developer to ensure it fall under budgeted plan
- PC4.** estimate total cost of ownership (Operating labour, software and hardware) for the implementation of robotic system in the organization on the basis of results of techno-commercial feasibility analysis

Pre and post support activities

To be competent, the user/individual on the job must be able to:

- PC5.** ensure the concerned department and system engineers are trained about usage and application before the system installation
- PC6.** arrange training for the users by system developers for easy access of automation system
- PC7.** ensure users get appropriate level of information access as per their usage requirement based sensitivity of the information

Process and scope of development

To be competent, the user/individual on the job must be able to:

- PC8.** analyse the ratio of automation implementation and study the process thoroughly
- PC9.** define the scope of development for the team in current process
- PC10.** define information flow among the team members related to the new technology of robotic automation

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to designing and maintaining network
- KU2.** relevant standards, procedures and policies related to robotic operations followed in the company
- KU3.** organizational procedure of interacting with vendors and others
- KU4.** Pre and post support activities need to perform
- KU5.** data safety and non-disclosers norms
- KU6.** cyber safety and work confidentiality good practices
- KU7.** data and folder standards
- KU8.** various check sheets and technical documents related to work
- KU9.** how to read visual controls, graphs etc.
- KU10.** robot anatomy and robot applications
- KU11.** various methods of information flow in the organisation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Collaboration with system developers</i>	15	15	-	8
PC1. interact with different vendors for developing the robotic automation system to meet all the information integration and information flow starting from conception of the idea till implementation of the robotic system	3	3	-	2
PC2. ensure that integrators/developers incorporate all the necessary requirement such what level of information access will be different department and different level of people as per requirement/hierarchy system	4	4	-	2
PC3. carry out the techno-commercial feasibility analysis with system developer to ensure it fall under budgeted plan	4	4	-	2
PC4. estimate total cost of ownership (Operating labour, software and hardware) for the implementation of robotic system in the organization on the basis of results of techno-commercial feasibility analysis	4	4	-	2
<i>Pre and post support activities</i>	15	15	-	6
PC5. ensure the concerned department and system engineers are trained about usage and application before the system installation	5	5	-	2
PC6. arrange training for the users by system developers for easy access of automation system	5	5	-	2
PC7. ensure users get appropriate level of information access as per their usage requirement based sensitivity of the information	5	5	-	2
<i>Process and scope of development</i>	10	10	-	6
PC8. analyse the ratio of automation implementation and study the process thoroughly	3	3	-	2
PC9. define the scope of development for the team in current process	3	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. define information flow among the team members related to the new technology of robotic automation	4	4	-	2
NOS Total	40	40	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8312
NOS Name	Liaison with vendors and other departments
Sector	Automotive
Sub-Sector	Research & Development
Occupation	Automotive Product Development
NSQF Level	7
Credits	4
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9810.Manage work and resources (Manufacturing)	50	30	-	20	100	15
ASC/N6438.Plan and design a network architecture using IIOT Solutions	40	40	0	20	100	20
ASC/N6447.Manage integration of machineries / Automation Systems in Healthy Communication via IIOT Sensor	40	40	0	20	100	20
ASC/N6448.Use Analytics, Edge & Cloud Computing Technologies to make Predictions on Production & Machineries	40	40	0	20	100	20
DGT/VSQ/N0104.Employability Skills (120 Hours)	20	30	-	-	50	10
ASC/N8312.Liaison with vendors and other departments	40	40	0	20	100	15
Total	230	220	0	100	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
I/O	Input/Output
IIOT	Industrial Internet of Things
I/O	Input/Output
IIOT	Industrial Internet of Things

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.